

Research Interests

Reinforcement Learning, Language Model Evaluation & Reward Modeling and **Multimodal LLM**

Work

Meta, Menlo Park, CA, USA May 2024 – Current
Research Scientist @ Monetization GenAI: Ads Creative Agent

- Designed and implemented an end-to-end **agentic RL framework** for ads image generation, combining policy optimization with multimodal reward signals; A/B tests demonstrated significant online gains including CTR improvements on a traditionally hard-to-optimize target; currently extending to **video ads generation agents**;
- Led end-to-end design of a **multimodal automated evaluation system** (Prompt → SFT → RL) to scalably assess LLM/VLM-generated ad creatives, with findings published in peer-reviewed venues ([arXiv:2603.11665](#), [arXiv:2602.11340](#));
- Built a **user preference prediction model** for automatic creative variant selection, achieving 40%+ save-rate lift vs. 10% for rule-based baselines in A/B tests.

Meta, Seattle, WA, USA June 2023 – Sept 2023
PhD Software Engineer Intern @ Ads Core ML: Privacy
Designed and Built the new ads model for Signal Loss Problem.

Google, Atlanta, GA, USA Oct 2022 – June 2023
Student Researcher @ Google Federated Assistant Team
Employed privacy-preserving machine learning (federated learning) and parameter-efficient transfer learning to solve the massive data requirement of ASR models and the substantial communication cost between servers and clients.

Meta, Seattle, WA, USA May 2022 – Aug 2022
PhD Software Engineer Intern @ Ads Core ML: Privacy
Worked on building Multi-Task Multi-Label (MTML) Hypergraph Neural Network (HGNN) for Signal Loss Problem.

SenseTime Research, Beijing, CHINA Feb 2019 – Jul 2019
Research Intern @ Smart City Group
Accelerating the neural network inference time in mobile devices with Neural Architecture Search (NAS) for stereo matching.

Education

Emory University, Atlanta, GA, USA Dec. 2019 – May. 2024
Ph.D., Department of Computer Science
Advisor: Prof. Carl J. Yang & Prof. Ying Guo

Tongji University, Shanghai, CHINA Sept 2014 – Jul 2018
B.E., School of Software Engineering
Advisor: Prof. Lin Zhang

Publications

* indicates equal contribution.

[ACL 2026] **Multi-Task Reinforcement Learning for Enhanced MLLM-as-a-Judge**
Junjie Wu, **Xuan Kan**, Shunwen Tan, Zihao He, Bo Pan, Kaitai Zhang
The 64th Annual Meeting of the Association for Computational Linguistics (ACL), Oral, 2026

[arXiv 2026] **Bi-Level Prompt Optimization for Multimodal LLM-as-a-Judge**

Bo Pan, **Xuan Kan**, Kaitai Zhang, Yan Yan, Shunwen Tan, Zihao He, Zixin Ding, Junjie Wu, Liang Zhao
arXiv:2602.11340, 2026

- [Submitted to MIA] **Atlas-free Brain Network Transformer**
Shuai Huang, **Xuan Kan**, Ames J Lah, Deqiang Qiu
arXiv:2510.03306, 2025
- [MedInfo 2025] **Multimodal Fusion of EHR in Structures and Semantics: Integrating Clinical Records and Notes with Hypergraph and LLM**
Hejie Cui, Xinyu Fang, Ran Xu, **Xuan Kan**, Joyce C Ho, Carl Yang
MedInfo, 2025
- [JBI 2025] **A Survey on Knowledge Graphs for Healthcare: Resources, Applications, and Promises**
Hejie Cui, Jiaying Lu, Shiyu Wang, Ran Xu, Wenjing Ma, Shaojun Yu, Yue Yu, **Xuan Kan**, Chen Ling, Joyce Ho
Journal of Biomedical Informatics, 2025
- [IEEE BHI 2024] **Multi-task Learning for Brain Network Analysis in the ABCD study**
Xuan Kan, Hejie Cui, Keqi Han, Ying Guo, Carl Yang
The IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2024
- [IEEE BHI 2024] **BrainODE: Dynamic Brain Signal Analysis via Graph-Aided Neural Ordinary Differential Equations**
Kaicao Han, Yi Yang, Zijie Huang, **Xuan Kan**, Yang Yang, Ying Guo, Lifang He, Liang Zhan, Yizhou Sun, Wei Wang, Carl Yang
The IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2024
- [ISBI 2024] **Convergent Brain Connectome Analysis with Brain Graph Transformers**
Keqi Han, **Xuan Kan**, Carl Yang
IEEE International Symposium on Biomedical Imaging (ISBI), 2024
- [arXiv 2024] **Parameter-Efficient Transfer Learning under Federated Learning for Automatic Speech Recognition**
Xuan Kan, Yonghui Xiao, Tien-Ju Yang, Nanxin Chen, Rajiv Mathews
arXiv preprint arXiv:2408.11873, 2024
- [ACL Findings] **Knowledge-Infused Prompting Improves Clinical Text Generation with Large Language Models**
Ran Xu, Hejie Cui, Yue Yu, **Xuan Kan**, Wenqi Shi, Yuchen Zhuang, Wei Jin, Joyce Ho, Carl Yang
The 62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024
- [NeurIPS 2024] **Open Visual Knowledge Extraction via Relation-oriented Multimodality Model Prompting**
Hejie Cui, Xinyu Fang, Zihan Zhang, Ran Xu, **Xuan Kan**, Xin Liu, Yue Yu, Manling Li, Yangqiu Song, Carl Yang
Advances in Neural Information Processing Systems (NeurIPS), 2024
- [IEEE BHI 2023] **Dynamic Brain Transformer with Multi-level Attention for Functional Brain Network Analysis**
Xuan Kan, Aodong Chen Gu, Hejie Cui, Carl Yang
The IEEE-EMBS International Conference on Biomedical and Health Informatics (IEEE BHI), 2023
- [KDD 2023] **R-Mixup: Riemannian Mixup for Biological Networks**
Xuan Kan^{*}, Zimu Li^{*}, Hejie Cui, Yue Yu, Ran Xu, Shaojun Yu, Zilong Zhang, Ying Guo, Carl Yang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2023
- [ISBI 2023] **Deep dag learning of effective brain connectivity for fmri analysis**
Yue Yu, **Xuan Kan**, Hejie Cui, Ran Xu, Yujia Zheng, Xiangchen Song, Yangqiao Zhu, Kun Zhang, Razieh Nabi, Ying Guo
IEEE 20th International Symposium on Biomedical Imaging (ISBI), 2023
- [ISBI 2023] **Transformer-based hierarchical clustering for brain network analysis**
Wei Dai, Hejie Cui, **Xuan Kan**, Ying Guo, Sanne van Rooij, Carl Yang
IEEE 20th International Symposium on Biomedical Imaging (ISBI), 2023

- [AAAI 2023] **Neighborhood-regularized Self-Training for Learning with Few Labels**
Ran Xu, Yue Yu, Hejie Cui, **Xuan Kan**, Yanqiao Zhu, Joyce C. Ho, Chao Zhang, Carl Yang
Proceedings of the AAAI International Conference on Artificial Intelligence (AAAI), **Oral**, 2023
- [IEEE TMI] **BrainGB: A Benchmark for Brain Network Analysis with Graph Neural Networks**
Hejie Cui, Wei Dai, Yanqiao Zhu, **Xuan Kan**, Antonio Aodong Chen Gu, Joshua Lukemire, Liang Zhan, Lifang He, Ying Guo, Carl Yang
IEEE Transactions on Medical Imaging (TMI), **IF: 11.037**, 2022
- [NeurIPS 2022] **Brain Network Transformer**
Xuan Kan, Wei Dai, Hejie Cui, Zilong Zhang, Ying Guo, Carl Yang
Proceedings of the Conference on Neural Information Processing Systems (NeurIPS), **Spotlight**, 2022
- [KDD 2022] **Data-Efficient Brain Connectome Analysis via Multi-Task Meta-Learning**
Yi Yang*, Yanqiao Zhu*, Hejie Cui, **Xuan Kan**, Lifang He, Ying Guo, Carl Yang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2022
- [MIDL 2022] **FBNetGen: Task-aware GNN-based fMRI Analysis via Functional Brain Network Generation**
Xuan Kan, Hejie Cui, Joshua Lukemire, Ying Guo, Carl Yang
International Conference on Medical Imaging with Deep Learning (MIDL), **Oral**, 2022
- [ECML 2021] **Zero-Shot Scene Graph Relation Prediction through Commonsense Knowledge Integration**
Xuan Kan, Hejie Cui, Carl Yang
The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2021
- [WWW 2019] **AutoTune: Autonomous Learning for Face Recognition in the Wild via Ambient Wireless Cues**
Xiaoxuan Lu*, **Xuan Kan***, Stefano Rosa, Bowen Du, Hongkai Wen, Andrew Markham, Niki Trigoni.
The Web Conference 2019, WWW, 2019
- [ICDCN 2019] **HydraDoctor: Real-time Liquids Intake Monitoring by Collaborative Sensing**
Bowen Du, Chris Xiaoxuan Lu, **Xuan Kan**, Kai Wu, Man Luo, Jianfeng Hou, Kai Li, Salil Kanhere, Yiran Shen, Hongkai Wen
Proceedings of the 20th International Conference on Distributed Computing and Networking (ICDCN), 2019
- [SenSys 2017] **VeriNet: User Verification on Smartwatches via Behavior Biometrics**
Chris Xiaoxuan Lu, Bowen Du, **Xuan Kan**, Hongkai Wen, Andrew Markham, Niki Trigoni
Proceedings of the First ACM Workshop on Mobile Crowdsensing Systems and Applications (CrowdSenSys), 2017

Scholarships, Awards & Honors

IEEE BHI Travel Award	2023
CIKM Registration Award	2022
Outstanding Graduates of Tongji University, China	2018
National Scholarship, The Ministry of Education of China	2017
Tongji University Programming Competition (Silver Prize)	2017
China Undergraduate Mathematical Contest in Modeling (Second Prize in National Level)	2016
Android Entrepreneurship Student Challenge by Google Inc. (Sliver Prize)	2016
Tongji University Scholarship for Outstanding Students	2015, 2016

Skills

Python, C++, PyTorch, LLM, Machine Learning, LM-as-a-Judge, Reward Model